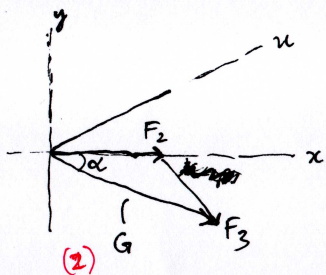


Problem 1-a



Let G be the magnitude of the resultant of $\vec{F}_2$ & $\vec{F}_3$.

~~By cosine law,~~

$$G^2 = F_2^2 + F_3^2 - 2F_2F_3 \cos[90^\circ + \tan^{-1}(3/5)]$$

0.5 kip 5 kip

$$\Rightarrow G \cong 5.274701771393112 \text{ kip} \quad (1)$$

By sine law, $\frac{F_3}{\sin \alpha} = \frac{G}{\sin[90^\circ + \tan^{-1}(0.6)]}$

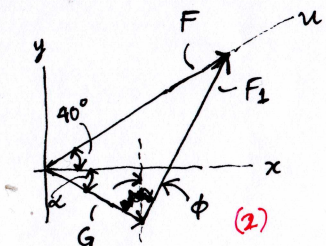
$$\Rightarrow \alpha \cong 54.37389567435551^\circ \quad (1)$$

Let F be the magnitude of the net force. ~~It is minimized~~

~~By inspection,~~ its minimum value is zero, i.e.,

when $F_1 = G \cong 5.274701771393112 \text{ kip} \quad (2)$

and $\phi \cong 270^\circ + \alpha \cong 324.3738956743555^\circ \quad (2)$



Problem 1-b

$$10\langle \cos(40^\circ), \sin(40^\circ) \rangle = F_1\langle \sin \phi, \cos \phi \rangle + 0.5\langle 1, 0 \rangle + 5\left\langle \frac{3}{34}, -\frac{5}{34} \right\rangle$$

This implies that

$$\begin{cases} F_1 \sin \phi = 10 \cos(40^\circ) - 0.5 - \frac{15}{\sqrt{34}} & (3) \\ F_1 \cos \phi = 10 \sin(40^\circ) + \frac{25}{\sqrt{34}} & (3) \end{cases}$$

So,

$$\tan \phi = \frac{10 \cos(40^\circ) - 0.5 - \frac{15}{\sqrt{34}}}{10 \sin(40^\circ) + \frac{25}{\sqrt{34}}} \Rightarrow \phi \cong 23.179058081189645^\circ \quad (2)$$

and therefore,

$$F_1 \cong 11.656241062400024 \text{ kips} \quad (2)$$

Problem 2

Let T_1, T_2 , and T_3 be the respective tensions in AB, AC, and AD, and

m be the mass of the pot. Observe that:

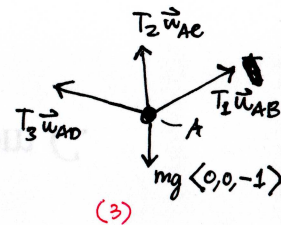
$$\vec{u}_{AB} = \langle 0, \sin(50^\circ), \cos(50^\circ) \rangle = \langle 0, \cos(40^\circ), \sin(40^\circ) \rangle$$

$$\vec{u}_{AC} = \langle -\sin(20^\circ), -\cos(20^\circ)\sin(40^\circ), \cos(20^\circ)\cos(40^\circ) \rangle$$

$$= \langle -\sin(20^\circ), -\cos(20^\circ)\cos(50^\circ), \cos(20^\circ)\sin(50^\circ) \rangle$$

$$\vec{u}_{AD} = \langle \sin(30^\circ), -\cos(30^\circ)\sin(40^\circ), \cos(30^\circ)\cos(40^\circ) \rangle$$

$$= \langle \sin(30^\circ), -\cos(30^\circ)\cos(50^\circ), \cos(30^\circ)\sin(50^\circ) \rangle$$



Static equilibrium (of the particle at A) requires that

$$T_1 \vec{u}_{AB} + T_2 \vec{u}_{AC} + T_3 \vec{u}_{AD} + mg\langle 0, 0, -1 \rangle = \langle 0, 0, 0 \rangle$$

which expands into the system of linear equations:

$$\begin{cases} 0T_1 - T_2 \sin(20^\circ) + T_3 \sin(30^\circ) + 0mg = 0 & (3) \\ T_1 \sin(50^\circ) - T_2 \frac{\cos(20^\circ)}{\sin(40^\circ)} - T_3 \frac{\cos(30^\circ)}{\sin(40^\circ)} + 0mg = 0 & (3) \\ T_1 \cos(50^\circ) + T_2 \frac{\cos(20^\circ)}{\cos(40^\circ)} + T_3 \frac{\cos(30^\circ)}{\cos(40^\circ)} - mg = 0 & (3) \end{cases}$$

The system can be solved if T_1, T_2 , and T_3 are pegged to their respective limits but one at a time. In other words, we analyze what happens if ~~a cable starts to~~ fail.

If AB fails, $T_1 = 60$, and

$$\begin{aligned} T_2 &\cong 46.67171481 < 50 \text{ N} & (\text{OK}) \\ T_3 &\cong 31.92533317 < 35 \text{ N} & (\text{OK}) \\ m &\cong 9.51838085 \text{ kg} \end{aligned} \quad (2)$$

If AC fails, $T_2 = 50$, and

$$\begin{aligned} T_1 &\cong 64.27876097 > 60 \text{ N} & (\text{NOT OK}) \\ T_3 &\cong 34.20201433 < 35 \text{ N} & (\text{OK}) \\ m &\cong 10.19716213 \text{ kg} \end{aligned} \quad (2)$$

If AD fails, $T_3 = 35$, and

$$\begin{aligned} T_1 &\cong 65.77848346 > 60 \text{ N} & (\text{NOT OK}) \\ T_2 &\cong 51.1665770 > 50 \text{ N} & (\text{NOT OK}) \\ m &\cong 10.43507763 \text{ kg} \end{aligned} \quad (2)$$

Therefore, the pot must weigh no more than 9.51838085 kg.

(2)

Problem 3

(a) The resultant force is $\left\langle \frac{1}{5}(0.1), -0.15 - 2.3 - 0.5 - \frac{3}{5}(0.1) \right\rangle$
 $= \langle 0.08, -2.71 \rangle$ kip or 2.711180554666177 kips, $\alpha = -88.30910273704966^\circ$

The couple moment about C is

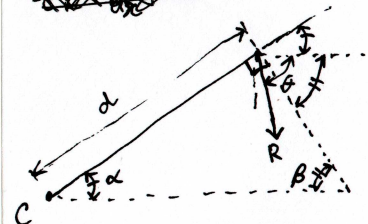
$$\frac{0.15(1.5)}{(1)} + \frac{2(3)}{(1)} + \frac{0.5(4.5)}{(1)} + \frac{\frac{3}{5}(0.1)(6)}{(1)} + \frac{\frac{1}{5}(0.1)(8)}{(1)} = 9.475 \text{ kip-ft, clockwise}$$

(b) In order for the resultant force to cause the same moment about C, whilst intersecting segment BC, the intersection must be above and to the right of C. Moreover, let the magnitude of the resultant be denoted by R and its direction clockwise from the x-axis be denoted by θ . By geometry,

$$\alpha = 90^\circ - \tan^{-1}(0.75) \quad \text{and} \quad \beta = \tan^{-1}(0.75)$$

Therefore, by the definition of moment (i.e., d as the moment arm),

$$R \cos(\theta - \beta) d = 9.475 \Rightarrow d \approx 5.606508875739645 \text{ ft}$$



Problem 4

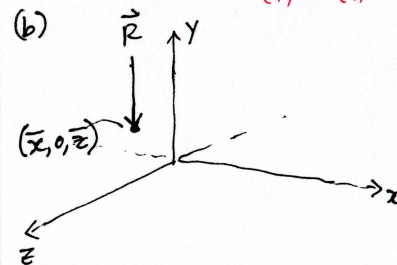
Observe that each column load is 3.85 m from O.

(a) The equivalent force-couple system at O comprises a resultant force $\vec{R}$ and a couple moment $\vec{M}_O$. By inspection,

$$\vec{R} = (150 + 110 + 200 + 40) \langle 0, -1, 0 \rangle \Rightarrow \vec{R} = 500 \langle 0, -1, 0 \rangle \text{ kN}$$

$$\vec{C} = \left\langle \underbrace{110(3.85) - 200(3.85)}_{\text{net moment (ccw) about x-axis}}, 0, \underbrace{150(3.85) - 40(3.85)}_{\text{net moment (ccw) about z-axis}} \right\rangle \text{ kN}$$

$$\Rightarrow \vec{C} = \langle -346.5, 0, 423.5 \rangle \text{ kN}\cdot\text{m}$$



In order for $\vec{R}$ to cause a clockwise moment about the x-axis and at the same time a counter-clockwise moment about the z-axis, it has to hit point $(\bar{x}, 0, \bar{z})$ where $\bar{x}, \bar{z} < 0$.

Hence, $|\vec{R}| |\bar{z}| = 346.5$
 and $\Rightarrow \bar{z} = -0.693 \text{ m}$
 $|\vec{R}| |\bar{x}| = 423.5$
 $\Rightarrow \bar{x} = -0.847 \text{ m}$

Therefore the single-force equivalent of the column loads intersects the mat at the point $(-0.847, 0, -0.693) \text{ m}$.